

The Unit-Transitivity Question in arXiv:math/0306365

A negative answer in arXiv:math/0411062

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Status: literature already answered (scoped negative answer). No new theorem is claimed. Tsirelson constructs a type II_1 Arveson system whose automorphism group is not transitive on its normalized units. This refutes the automorphism-transitivity statement explicitly left open in Liebscher's Note 3.11.

1 The source question

For a product system $\mathcal{E}$ and a unit u , Liebscher associates a stationary factorizing random-set measure type $\mathcal{M}^{\mathcal{E},u}$. Note 3.11 asks first whether

$$\mathcal{M}^{\mathcal{E},u} = \mathcal{M}^{\mathcal{E},v} \quad (u, v \text{ normalized units}),$$

so that this construction would be independent of the chosen unit. The note then records a sufficient mechanism: the equality follows whenever some automorphism θ of $\mathcal{E}$ satisfies

$$v_t = \theta_t u_t \quad (t > 0).$$

It explicitly calls the universal existence of such an automorphism an open problem.

Note 3.10 We prove in Corollary 6.2 below that $\{\mu_\eta : \eta \text{ faithful}\}$ is already a measure type, i.e. it is equal to $\mathcal{M}^{\mathcal{E}, \mathcal{U}}$.

We present now two simple examples.

Example 3.6 If $\mathcal{E}$ is of type III, i.e. it has no unit, we know $P_{s,t}^{\mathcal{U}} = 0$ for all $(s, t) \in I_{0,1}$. Thus Example 3.4 shows $\mu_\eta^{\mathcal{U}} = \delta_{[0,1]}$ and $\mathcal{M}^{\mathcal{E}, \mathcal{U}} = \{\delta_{\{[0,1]\}}\}$.

Example 3.7 Consider $\mathcal{E} = I(\mathcal{K})$ which is of type I. Thus $P_{0,1}^{\Gamma(\mathcal{K}), \mathcal{U}} = \mathbb{1}$ and Example 3.5 yields $\mathcal{M}^{\mathcal{E}, \mathcal{U}} = \{\delta_\emptyset\}$.

3.4 Measure Types Related to Units

It is clear, that for the first part of Theorem 2 the family $(P_{s,t})_{(s,t) \in I_{0,1}}$ need not be associated with $\mathcal{E}^{\mathcal{U}}$. So we obtain in a similar, but due to $P_{0,1}^u = \text{Pr}_{u_1} \neq 0$ more simple, fashion

Proposition 3.6 For every product system $\mathcal{E}$ and every unit $u \in \mathcal{U}(\mathcal{E})$ the measures μ_η^u defined via Theorem 1 from the system $(P_{s,t}^u)_{(s,t) \in I_{0,1}}$ with varying faithful normal state η on $\mathcal{B}(\mathcal{E}_1)$ belong to a unique measure type $\mathcal{M}^{\mathcal{E}, u}$ fulfilling (3.15) and (3.16) ■

Note 3.11 We would like to have, like for the product subsystem $\mathcal{E}^{\mathcal{U}}$, that $\mathcal{M}^{\mathcal{E}, u}$ is an invariant of the product system $\mathcal{E}$. So the main problem is whether $\mathcal{M}^{\mathcal{E}, u}$ is a well-defined invariant, i.e. whether for two normalized units u, v the measure types $\mathcal{M}^{\mathcal{E}, u}$ and $\mathcal{M}^{\mathcal{E}, v}$ are equal. Copying the relevant part of the proof of Theorem 2 gives us only $\mathcal{M}^{\mathcal{E}, u} = \mathcal{M}^{\mathcal{E}', \theta_1 u}$ if θ is an isomorphism between $\mathcal{E}$ and $\mathcal{E}'$. Thus $\mathcal{M}^{\mathcal{E}, u} = \mathcal{M}^{\mathcal{E}, v}$ would result for all $u, v \in \mathcal{U}(\mathcal{E})$ if there were an automorphism of $\mathcal{E}$ such that $v_t = \theta_t u_t$. Whether the latter is true is an open problem until now [15], although we have strong indications (see Remark 10.2 and Proposition 3.9) that the answer is affirmative.

To derive the connection between the measure types $\mathcal{M}^{\mathcal{E}, u}$ and $\mathcal{M}^{\mathcal{E}, \mathcal{U}}$ we need to collect more properties of units. It is easy to prove (cf. [4] Theorem 4.1) that for two units $u, v \in \mathcal{U}(\mathcal{E})$ there is some $\gamma(u, v) \in \mathbb{C}$ such that

$$\langle u_t, v_t \rangle_{\mathcal{E}_t} = e^{-\gamma(u, v)t}, \quad (t \in \mathbb{R}_+). \quad (3.19)$$

Thereby, $\gamma : \mathcal{U}(\mathcal{E}) \times \mathcal{U}(\mathcal{E}) \rightarrow \mathbb{C}$ is a conditionally positive kernel and called covariance function [4].

Poisson processes Π_v are distributions of (finite) random closed sets defined by

$$\Pi_v = e^{-v([0,1])} \left(\delta_\emptyset + \sum_{n \in \mathbb{N}} \frac{1}{n!} \int v^n(dt_1, \dots, dt_n) \delta_{\{t_1, \dots, t_n\}} \right). \quad (3.20)$$

where v is a finite measure on $[0, 1]$ with $v(\{t\}) = 0 \forall t$. Observe that $\Pi_v(\{Z : Z \cap G = \emptyset\}) = e^{-v(G)}$ for all (open) sets $G \subset [0, 1]$ which is characteristic for Π_v by the Choquet theorem [39, Theorem 2-2-1]. Observe that this theorem shows too that Π_v is factorizing and stationary if $v = \ell$. It is not hard to prove from (3.20) that it is quasifactorizing for all $v \sim \ell$.

Lemma 3.7 Let u, v be two units with $\|v_1\| = 1$. Then the measure μ_η^u associated with the projection family $(P_{s,t}^u)_{(s,t) \in I_{0,1}}$ and the pure normal state $\eta(a) = \text{tr Pr}_{v_1} a \forall a \in \mathcal{B}(\mathcal{E}_1)$ by Theorem 1 is the Poisson process $\Pi_{2\Re \gamma(u, v)\ell}$.

Figure 1: Liebscher, arXiv:math/0306365, source PDF page 20. Note 3.11 contains both the measure-type question and the unit-transitivity route.

2 The later counterexample

Tsirelson's paper is directed at precisely this transitivity conjecture. Its Theorem 1.10 states:

There exists an Arveson system of type II_1 such that the action of its automorphism group on the set of normalized units is not transitive.

Thus there are a product system $\mathcal{E}$ and normalized units u, v for which no automorphism θ has $v_t = \theta_t u_t$ for every $t > 0$. This is a literal counterexample to the universal assertion in the final sentence of Liebscher's Note 3.11.

where $f \in L_2(\mathbb{R}^n)$ (or rather, the relevant part of $\mathbb{R}^n$). One may say that $\Theta^{\text{rotat}(\lambda)}$ just multiplies each dB_s by $e^{i\lambda}$.

Combining shifts and rotations we get all motions of $\mathbb{C}$. Accordingly, all automorphisms of $(H_t^{\text{white}})_t$ are combinations of $\Theta^{\text{shift}(i\lambda)}$, $\Theta^{\text{shift}(\lambda)}$, $\Theta^{\text{rotat}(\lambda)}$ and $\Theta^{\text{trivial}(\lambda)}$. More generally, the N -dimensional Brownian motion leads to the (unique up to isomorphism) Arveson system of type I_N and motions of $\mathbb{C}^N$. We need $N = 1$ only; $(H_t^{\text{white}})_t$ is the Arveson system of type I_1 .

Some noises are constructed as extensions of the white noise,

$$(1.8) \quad \mathcal{F}_{s,t} \supset \mathcal{F}_{s,t}^{\text{white}}$$

(also T_h conforms to T_h^{white}). More exactly, it means that $B_t \in L_2(\mathcal{F})$ are given such that $B_t - B_s$ is $\mathcal{F}_{s,t}$ -measurable for $-\infty < s < t < \infty$, and $B_t - B_s \sim N(0, t-s)$ (that is, the random variable $(t-s)^{-1/2}(B_t - B_s)$ has the standard normal distribution), and $B_0 = 0$, and $(B_t - B_s) \circ T_h = B_{t+h} - B_{s+h}$. Such $(B_t)_t$ may be called a Brownian motion adapted to the given noise. Then, of course, by $\mathcal{F}_{s,t}^{\text{white}}$ we mean the sub- σ -field generated by $B_u - B_s$ for all $u \in (s, t)$. The Arveson system $(H_t)_t$, $H_t = L_2(\mathcal{F}_{0,t})$, is an extension of the type I_1 system $(H_t^{\text{white}})_t$, $H_t^{\text{white}} = L_2(\mathcal{F}_{0,t}^{\text{white}})$,

$$(1.9) \quad H_t \supset H_t^{\text{white}}.$$

All units of $(H_t^{\text{white}})_t$ are also units of $(H_t)_t$. It may happen that $(H_t)_t$ admits no other units even though $\mathcal{F}_{s,t} \neq \mathcal{F}_{s,t}^{\text{white}}$, $H_t \neq H_t^{\text{white}}$. Then $(H_t)_t$ is of type II (units generate a nontrivial, proper subsystem), namely, of type II_1 ; $(H_t^{\text{white}})_t$ is the classical part of $(H_t)_t$, and the white noise is the classical part of the given noise. The automorphisms $\Theta^{\text{trivial}(\lambda)}$ and $\Theta^{\text{shift}(i\lambda)}$ for $\lambda \in \mathbb{R}$ can be extended naturally from the classical part to the whole system (which does not exclude other possible extensions). For $\Theta^{\text{shift}(\lambda)}$ and $\Theta^{\text{rotat}(\lambda)}$ we have no evident extension. Moreover, these automorphisms need not have any extensions, as will be proved.

Two examples found by Warren [11], [12] are 'the noise of splitting' and 'the noise of stickiness'; see also [13] and [9] Sect. 2]. For the noise of splitting the gauge group restricted to the classical part covers all shifts of $\mathbb{C}$ (but only trivial rotations [10]), thus, it acts transitively on $\mathbb{C}$, therefore, on normalized units as well.

A new (third) example is introduced in Sect. 10 for proving the main result formulated as follows.

1.10 Theorem. There exists an Arveson system of type II_1 such that the action of the group of automorphisms on the set of normalized units is not transitive.

3 Scope and logical crosswalk

It is important not to overstate what the counterexample proves. The source passage contains two statements:

1. unit-independence of $\mathcal{M}^{\mathcal{E},u}$;
2. transitivity of $\text{Aut}(\mathcal{E})$ on normalized units.

Liebscher proves that (2) would imply (1). Tsirelson proves that (2) is false. The converse implication was not asserted, so the counterexample does not by itself show that the two measure types in (1) differ. Accordingly this packet claims a complete negative answer only to the automorphism-transitivity subquestion, not to the unit-independence question.

The matching quantifiers are exact:

Source assertion	Later result
For every product system and every pair of normalized units, an automorphism maps one unit to the other.	A type II_1 product system has two normalized units in distinct automorphism orbits.

4 Provenance and verification

The search began with the run’s local registry, attempt, solution, and proof-gap indexes; none contained arXiv:math/0306365 or this question. A bounded title-and-keyword search then found arXiv:math/0411062. The displayed question and theorem were checked directly in the two primary PDFs, rather than inferred from abstracts. Both PDFs are included in this packet.

References

- [1] V. Liebscher, *Random Sets and Invariants for (Type II) Continuous Tensor Product Systems of Hilbert Spaces*, arXiv:math/0306365; Mem. Amer. Math. Soc. 199 (2009), no. 930.
- [2] B. Tsirelson, *On automorphisms of type II Arveson systems (probabilistic approach)*, arXiv:math/0411062; New York J. Math. 14 (2008), 539–576.